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MILITARY SPECIFICATION

# CLEANING COMPOUND, SOLVENT (FOR BORE OF SMALL ARMS AND AUTOMATIC AIRCRAFT WEAPONS)

*This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.*

## 1. SCOPE

1.1 This specification covers one type of bore cleaner for use in the removal of primer salts and as a temporary preservative for metallic items.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

### SPECIFICATIONS

FEDERAL

O-M-232 — Methanol (Methyl Alcohol).

QQ-S-640 — Steel, Carbon; Sheet and Strip.

RR-S-366 — Sieves, Standard for Testing Purposes.

TT-N-95 — Naphtha, Aliphatic.

PPP-C-96 — Cans, Metal, 28 Gage and Lighter.

MILITARY

JAN-H-792 — Humidity Cabinet; Operation of.

### STANDARDS

FEDERAL

Federal test Method Standard No. 791 Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

MILITARY

MIL-STD-290 — Packaging, Packing, and Marking of Petroleum and Related Products.

(Copies of specifications, standards, drawings and other publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to

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the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS PUBLICATIONS

ASTM Standard on Petroleum Products and Lubricants.

ASTM Manual on Measurement and Sampling of Petroleum and Petroleum Products.

(Copies of ASTM Publications may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

## 3. REQUIREMENTS

**3.1 Qualification.** The bore cleaner furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein (see 4.5.1 and 6.6), and has been listed on or approved for listing on the applicable qualified qualified products list.

**3.2 Material.** The bore cleaner shall be made of such materials as shall meet all requirements of this specification. The cleaner shall contain no ingredients which may be injurious to personnel using it under reasonable safety precautions.

**3.3 Physical requirements.** The cleaner shall have the following physical requirements (see 4.6):

Flash Point, degrees Fahrenheit (°F.), Minimum 120

Viscosity, centistakes (cs.)

@100°F., Max .....15

@—20°F., Max .....1500

**3.4 Stability.**

3.4.1 *Freezing.* The cleaner shall show no separation after storage throughout a temperature range of —65°F. to 77°F. (see 4.7.1).

3.4.2 *Heating.* The cleaner shall show no separation after storage at 150°F. (see 4.7.2).

**3.5 Residue and fluidity at low temperature.** The residue remaining after evaporation at 130°F. shall be free from solid particles and not tacky and shall permit movement at —65°F. when tested as specified in 4.8.

**3.6 Appearance.** The cleaner shall be clear and transparent when viewed by transmitted light (see 4.9).

**3.7 Odor.** The cleaner shall be free from disagreeable and offensive odors when tested as specified in 4.10. (An obvious strongly offensive odor shall be cause for immediate rejection without conducting the test.)

**3.8 Corrosion protection.**

3.8.1 *Performance (primer salts removal).* The cleaner shall remove primer salts and prevent rusting. The presence of a discoloration shall not be cause for rejection (see 4.11.1).

3.8.2 *Humidity cabinet.* The cleaner shall be capable of protecting parts against corrosion during continuous condensation on parts at temperatures up to 120°F. (see 4.11.2).

3.8.3 *Water displacement and water stability.* The cleaner, after storage in contact with water, shall satisfactorily displace water as evidenced by the absence of rust, mottling, or other abnormal surface stains on the test specimens (see 4.11.3).

3.8.4 *Container corrosion.* The cleaner shall not produce visually evident corrosive effects or a weight gain or loss in excess of the following (see 4.11.4):

| *Container finish* | *Milligrams per square centimeter* |
|---|---|
| Electrolytic tinplate — 0.25 lbs per base box ..... | 0.3 |
| Electrolytic tinplate — 0.50 lbs per base box ..... | 0.3 |

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Electrolytic tinplate —
0.75 lbs per base box ..... 0.3

Hot dipped tinplate —
1.25 lbs per base box ..... 0.4

**3.9 Toxicity.** When used for its intended purpose, the cleaner shall have no adverse effect on the health of personnel (see 6.5).

## 4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Lot.** An inspection lot shall consist of cleaner of a single batch or composition, manufactured under essentially the same conditions, and presented for inspection at the same time. However, a lot shall not exceed 5,000 gallons in any case.

4.3 **Sampling.**

4.3.1 *Sampling material for testing.* Each lot of material shall be sampled in accordance with Method 8001 of Federal Test Method Standard No. 791. Two separate one-gallon samples shall be taken from different containers in each lot. The samples shall be placed in clean, dry, metal containers. One container shall be sent to the test laboratory specified in the contract. The remaining container shall be retained by the Government inspector until notification is received that the sample is acceptable, or that the remaining sample is required for testing by the laboratory. The containers shall be marked as follows:

Sample for acceptance tests.

This specification number.

Name of manufacturer.

Manufacturer's code number.

Batch or lot number.

Date of sampling.

4.3.2 *Qualification test samples.* Qualification samples shall consist of two one-gallon containers of the cleaner.

4.3.2.1 The qualification samples shall be accompanied by test data indicating results of all tests except the container corrosion test.

4.4 **Inspection.**

4.4.1 *Inspection of material.* Inspection shall be in accordance with Method 9601 of Federal Test Method Standard No. 791.

4.4.2 *Inspection of packaging.* The containers shall be inspected for fill, closure, evidence of leakage, and marking to determine compliance with Standard MIL–STD–290.

4.5 **Classification of tests.**

4.5.1 *Qualification tests.* Qualification tests shall consist of all the tests of this specification.

4.5.2 *Acceptance tests.* Acceptance tests shall consist of all the tests of this specification except the odor test (see 4.10) and the container corrosion test (see 4.11.4).

4.6 **Physical and chemical tests.** Physical and chemical tests shall be conducted in accordance with Federal Test Method Standard No. 791 as follows:

(a) Flash Point ..... Method 1102

(b) Viscosity ..... Method 305

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4.7 Stability.

4.7.1 *Freezing.* A sample of the cleaner shall be placed in a pour point tube, stoppered, and subjected to —65 ± 5°F. for 16 hours followed by 8 hours at 77 ± 5°F. This cycle shall be repeated for a total of three times. Following this procedure the sample shall be inverted once, permitted to stand at 77 ± 5°F. for 16 hours, and then examined for separation. Any evidence of separation in the sample shall be cause for rejection.

4.7.2 *Heating.* A sample of the cleaner shall be placed in a pour point tube, loosely stoppered, and subjected to 150°F. for 30 minutes. Upon completion of the heating period the sample is permitted to cool to 77 ± 5°F. and examined for separation. Any evidence of separation in the sample shall be cause for rejection.

4.8 Residue and fluidity at low temperature. Twenty-five milliliters (ml.) of the cleaner shall be placed in a flat bottom petri dish, approximately 10 centimeters (cm.) in diameter, and heated in an oven maintained at 130 ± 5°F. for 40 hours. At the end of the heating period, the sample shall be cooled to 77 ± 5°F. and examined for tackiness by touching the residue. A visual examination shall be made for solid particles. A sufficient quantity of the residue shall be placed upon the end of a glass slide approximately 1 by 3 inches. Another slide shall be placed over the residue so that an overlap of approximately one square inch is formed. A one kilogram weight is placed on the test area for 30 seconds. The weight is then removed and the slides subjected to —65 ± 5°F. for 24 hours. While at a temperature of —65°F. the two slides shall be separated by hand using a sliding action. If no movement is detected within a period of one second the material shall be rejected.

4.9 Appearance. A pour point tube shall be filled to the mark with the sample to be tested. The sample is then viewed by transmitted light and shall be clear and transparent.

4.10 Odor.

4.10.1 *Procedure.* Twenty ml. of the cleaner is placed in a weighing bottle, 70 mm I.D. and 30 mm in height, sealed with a ground glass stopper, and maintained in an oven at 100 ± 5°F. for 30 minutes. Immediately after removal from the oven, the container is transferred to a draft free area, the stopper is removed, and the vapors rated by each of three operators, as to being agreeable or offensive (see 4.10.2) according to the following procedure.

(a) The operator should not be more than 2-3 inches from the weighing bottle when removing the stopper and smelling the vapors.

(*Note.* In case strong odors are present it is recommended that the operator partially fill his lungs with air prior to removing stopper to only permit exposure to sufficient vapors for odor detection.)

(b) The operator should be exposed to the vapors for 15 seconds after removing stopper, breathing carefully.

(c) The operator shall determine the length of time he can sense or notice the odor after completion of the 15 second exposure period.

4.10.2 *Rating the odor.* Each operator shall rate the odor as follows as to category and/or type:

Category 1 — *Agreeable*

Type A — Pleasant.

Type B — No odor.

Type C — Neutral odor.

Category 2 — *Offensive*

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Type A — Nauseous.

Type B — Repellent.

Type C — Burning.

Type D — Strongly penetrating.

Type E — Drying.

Type F — Lachrymatory.

Type G — Persistence after 15 seconds exposure.

Type H — Dizziness.

Type I — Sneezing.

Type J — Other adverse effects.

4.10.3 *Criterion for failure.* If any 2 of 3 operators rate the odor as offensive, the material shall be considered unsatisfactory for use and rejected.

4.11 Corrosion protection.

4.11.1 *Performance (primer salts removal).*

4.11.1.1 *Test panels.* The test panels shall be 2 by 4 by ⅛ inch; low carbon, cold-rolled 1020 steel conforming to Specification QQ–S–640. Badly rusted stock shall not be used for making test panels. The edges of the panels shall be rounded and suspension holes reamed in accordance with Specification JAN–H–792, prior to cleaning.

4.11.1.2 *Cleaning test panels.* The test surfaces of the panels shall be cleaned with naphtha conforming to Specification TT–N–95, and methanol conforming to Specification O–M–232, Grade A, as follows:

(a) While cleaning the test panels they shall be handled with hooks or forceps at all times. All precautions shall be taken to guard against impurities on the test panels by avoiding contact with any type of contaminated surfaces. The utensils and solvents used must be cleaned and free from contamination.

(b) The solvents shall be maintained at a temperature high enough to keep the temperature of the panels above the dew point during handling operations when they are not submerged in solvent or stored in a desiccator.

(c) Wipe the surfaces clean with solvent soaked rags and scrub with surgical gauze swabs in a beaker of hot naphtha.

(d) Rinse in a beaker of hot methanol. Air dry the panels and store in a desiccator until further processing.

4.11.1.3 *Sandblasting test panels.* The unnumbered side of the test panels shall be sandblasted to a fresh, uniformly abraded surface with clean, white, dry, sharp sand, of a size that will allow all of it to pass through a number 10 sieve, a minimum of 90 percent to pass through a number 20 sieve, and not more than 10 percent to pass through a number 50 sieve. The size designation of all sieves shall conform to Specification RR–S–366. Immediately after sandblasting, the panels shall be placed into a container of anhydrous methanol. Remaining residue and contamination shall be removed by holding the panels in a rack at 20° from the vertical and spraying downward with naphtha. Flush the test surface, then the back of the panel and the test surface again. The panels shall then be rinsed in hot naphtha and hot methanol. After the panels are dry they shall be stored in a desiccator and used the same day as prepared.

4.11.1.4 *Test procedure.* Three test panels shall have a Cal. .30 primed, (corrosive) empty cartridge case fired at the center of each (see 6.3). The primed case shall be held in a rigid manner at 45° obliquity with a distance of one inch between the cartridge and the test panel. The panels shall then be slushed in approximately 800 ml. of the

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cleaner under test for 2 minutes. The panels shall then be suspended by stainless steel or monel hooks and exposed in the humidity cabinet specified in Specification JAN–H–792 for 3 days. The slushing of the panels in the test cleaner shall be accomplished mechanically by use of a reciprocal stroking machine such as shown in Figure 1. The panels shall be mounted in the test holder in a vertical position, with one on each side of the holder, so that the contaminated side of the panels is in a plane perpendicular to the path of the slushing motion. The length of the stroke shall be 2 ± ¼ inch and the slushing shall be conducted at 30 ± 1 cycles per minute. After exposure of the panels in the humidity cabinet for 3 days, the panels shall be examined in the significant area of the panels as defined in Specification JAN–H–792. At least two of the panels shall be free of corrosion and the third panel shall have no more than three corrosion dots, none of which exceed one millimeter in diameter. Corrosion in excess of this shall be cause for rejection.

4.11.2 *Humidity cabinet.*

4.11.2.1 *Test panels.* Three test panels for the humidity cabinet test shall be of the same size and material as specified in 4.11.1.1, and cleaned and sandblasted as specified in 4.11.1.2 and 4.11.1.3.

4.11.2.2 *Test procedure.* The test panels shall be dipped in the test cleaner at a temperature of 77 ± 5°F. and agitated gently for one minute. The panels shall then be suspended by means of stainless steel or monel hooks and drained for two hours at that temperature. The panels shall then be subjected to 7 days of the humidity cabinet test specified in Specification JAN–H–792. At the end of the 7 day exposure period, the panels shall be removed, decoated with naphtha and examined in the significant area of the panels as defined in Specification JAN–H–792. At least two of the panels shall be free of corrosion and the third panel shall have no more than three corrosion dots, none of which exceed one millimeter in diameter. Corrosion in excess of this amount shall be cause for rejection.

4.11.3 *Water displacement and water stability.*

4.11.3.1 *Test panels.* Three panels for this test shall be 2 by 3 by 1/16 inch of the same material specified in 4.11.1.1, and cleaned and sandblasted as specified in 4.11.1.2 and 4.11.1.3.

4.11.3.2 *Preparation of compound.* Fifty ml. of the test compound shall be placed in a 125 ml. Erlenmeyer flask, 5 ml. of distilled water added, and the mixture thoroughly shaken. The stoppered flask shall be stored in an oven at 130 ± 5°F. for 16 hours and then cooled to 77 ± 5°F. for testing.

4.11.3.3 *Test procedure.* Three test panels shall be dipped into distilled water momentarily and drained in a vertical position for not more than five seconds, with the bottom edge in contact with absorbent paper. The panels shall be immersed horizontally and without agitation for 15 seconds in a petri or evaporating dish containing 50 ml. of the test compound. After draining momentarily, the panels shall be placed in a static humidity chamber, (for example, a desiccator body containing some distilled water), for one hour at 77 ± 5°F. At the completion of the static humidity exposure the panels shall be decoated with naphtha and examined. The presence of rust, mottling or other abnormal surface stains shall be cause for rejection.

4.11.4 *Container corrosion.*

4.11.4.1 *Test containers.* This test shall be conducted utilizing the following containers, round, 6 ounce capacity, conforming to PPP–C–96:

(a) Electrolytic tinplate — 0.25, 0.50 and 0.75 lbs per case box.

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(b) Hot dipped tinplate — 1.25 lbs per base box.

4.11.4.2 *Preparation of test containers.* The containers shall be rinsed in hot naphtha followed by a rinse in hot methyl alcohol. The containers shall be permitted to cool to 75 ± 5°F. prior to further processing. Avoid touching containers with the hands after the cleaning process.

4.11.4.3 *Test procedure.* Weigh each container and add 110 ml. of the compound under test. The lids shall be securely attached and the containers placed in an oven at 130 ± 5° for 7 days. Upon completion of the test, the cleaner shall be removed and the containers rinsed in hot naphtha and hot methyl alcohol. After cooling to 75 ± 5°F. reweigh the containers and calculate the change in weight in milligrams per square centimeter (see 6.4).

## 5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.** The bore cleaner shall be furnished in 2-ounce, 6-ounce, 1-quart, 1-gallon screwtop cans, 5-gallon pails, and 55-gallon drums as specified by the procuring activity (see 6.2). Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with MIL–STD–290. The level of packaging shall be level A or C, and the level of packing shall be level A, B or C, as specified (see 6.2).

5.2 **Marking.** The unit shipping containers shall be marked for shipment in accordance with Standard MIL–STD–290.

## 6. NOTES

6.1 **Intended use.** The bore cleaner covered by this specification is a highly penetrating, mobile liquid and is intended for use in cleaning the bores of small arms and automatic aircraft weapons. The material provides a temporary rust resistant coating for the cleaned surface.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title and number of this specification.
(b) Level of packaging and level of packing required.
(c) Size and type of container.
(d) Any special marking required.
(e) Quantity desired.

6.3 **Primed cartridge cases.** Primed (corrosive), empty cartridge cases required for the performance test (par. 4.11.1.4) may be obtained from the Contracting Officer, Raritan Arsenal, Metuchen, New Jersey.

6.4 The surface area of the containers can be determined by utilizing the following formulas:

Area of circle $= \pi r^2$

Area of cylinder $= 2 \pi rh$

where r (centimeters) = radius of bottom of container
and h (centimeters) = height of liquid in container

The total surface area can be obtained by combining the individual values.

6.5 Any questions raised regarding toxicity should be referred by the procuring agency to the departmental medical authority. In the case of Army procurement, the Surgeon General will act as advisor to the procuring agency.

6.6 **Qualification.** With respect to products requiring qualification, awards will be made only for such products as have, prior to bid opening date, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to

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be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is the Commanding Officer, Rock Island Arsenal, Rock Island, Ill., and information pertaining to qualification of products may be obtained from that activity.

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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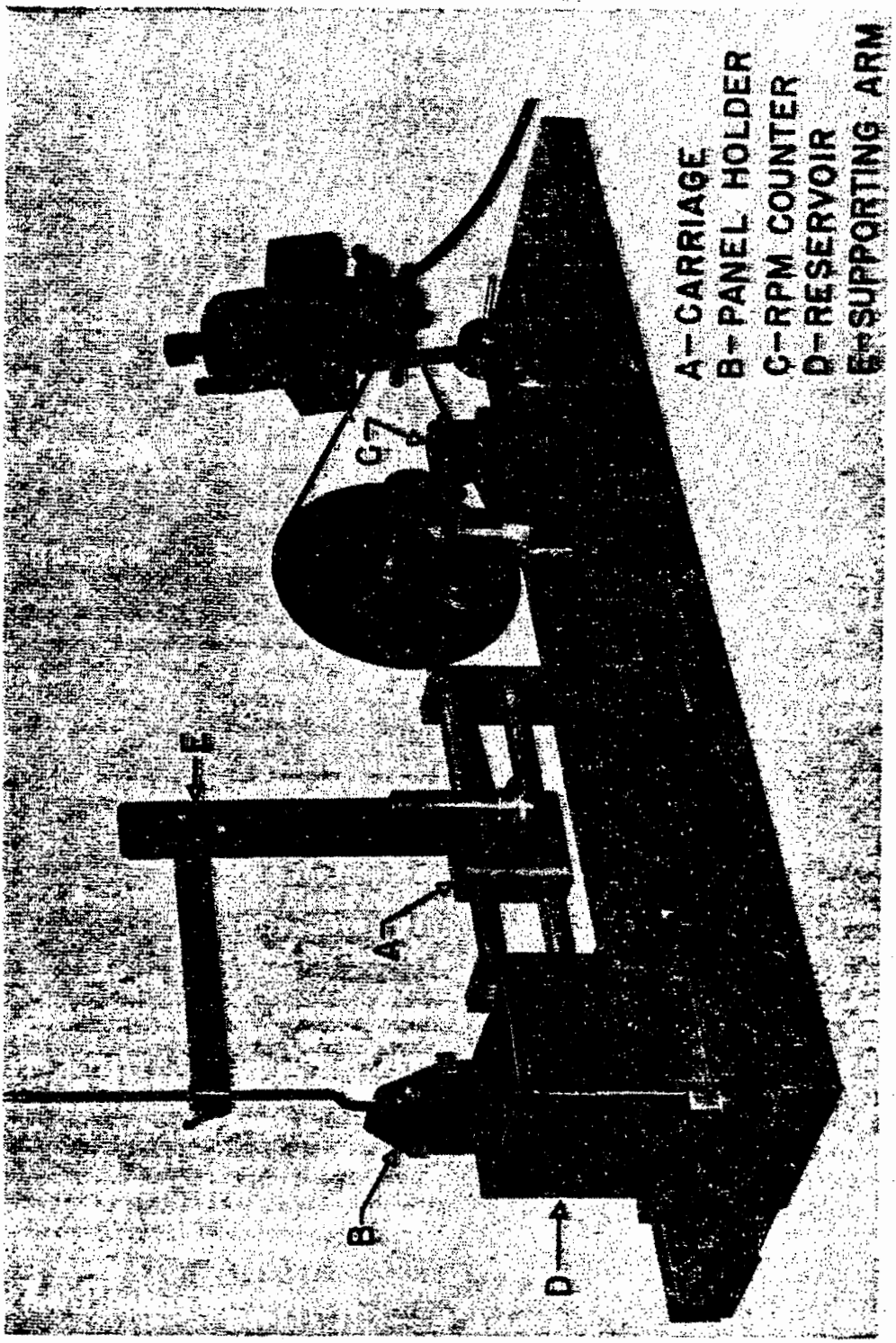



FIGURE 1

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